



Materials Science and Engineering: A

Volume 957, April 2026, 149978

Microstructure and mechanical properties of MIG-welded 6061-T6 aluminum alloy joints with 6061–TiC filler wire: synergistic effects of TiC nanoparticles and post-weld heat treatment



Wenbin Jiang ^{a b}, Chenxu Zhao ^{a c}, Yao Huang ^{a b}, Yanjian Ding ^{a b}, Liang Qi ^{a b d} , Jiqiang Chen ^{a b d} , Zhou Li ^{a b}

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<https://doi.org/10.1016/j.msea.2026.149978>

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Abstract

Welding 6000-series aluminum alloy with 6000-series filler wire maintains the same chemical composition between the weld zone (WZ) and the base material (BM), which is beneficial for subsequent material recycling, but it also makes the joint more susceptible to hot cracking during the welding process. In this study, 6061 aluminum alloy was welded by the metal inert gas (MIG) process using 6061–TiC filler wire containing 1.5wt% TiC nanoparticles, and the resulting joints were compared with those produced using a conventional 4043 filler wire. The coupled effects of nanoparticle addition and post-weld heat treatment (PWHT) on the microstructure and mechanical properties of 6061 joints were systematically assessed. The results indicate that

TiC nanoparticles refine the solidified microstructure in the WZ, transforming coarse columnar dendrites (83.4 μm) into fine equiaxed grains (13.7 μm), thereby inhibiting hot crack formation. At the same time, a weak texture with random orientations is formed, which is conducive to uniform plastic deformation. The $\text{Al}_{18}\text{Mg}_3\text{Ti}_2$ and Al_3Ti phases formed during welding facilitated the precipitation of high-density nanoscale β'' strengthening phases during the PWHT process. The resulting microstructure improves the strength–ductility balance in the 6061 joints. The tensile strength of the joint is 281.7 MPa, the elongation is 17.0%, and its elongation is 59% higher than that of the joint produced using 4043 filler wire. These findings indicate that nanoparticle-assisted solidification refinement enhances resistance to weld hot cracking and improves ductility, while PWHT-induced β'' precipitation increases strength. This welding approach facilitates high-performance welding of 6000-series aluminum alloys and enhances recyclability, contributing to sustainable manufacturing.



Keywords

6061 aluminum alloy; TiC nanoparticles; PWHT; Grain refinement; β'' precipitation

1. Introduction

The global push toward carbon peaking and carbon neutrality has positioned lightweight transportation as a key route to energy saving and emissions reduction [1,2]. Within aluminum alloys, the 6000-series, particularly 6061, offers attractive combinations of specific strength, formability, and corrosion resistance for structural components and battery enclosures [3], [4], [5], [6], [7]. Several fusion welding methods can be employed to join Al-alloys, such as laser or electron beam welding [8,9]. The metal inert gas (MIG) welding process, which is widely employed in industrial applications to join steels [10], [11], [12], presents some difficulties in joining Al-alloys. Porosity formation is an intrinsic problem in fusion welding of Al-alloys, including MIG welding [13,14]. Although solid state friction stir welding (FSW) can be used to eliminate porosity formation, the loss of strength in the weld region, i.e., HAZ, cannot be avoided, particularly in 6061 alloy [15,16]. Furthermore, 6061 aluminum alloy is susceptible to hot cracking during fusion welding owing to a wide solidification temperature range and inadequate interdendritic feeding [17],

[18], [19]]. Controlled, filler-free lap-welding tests confirmed the alloy's higher susceptibility to cracking than that of 2024 and 7075 alloys [20].

Strategies to mitigate hot-cracking susceptibility generally fall into two categories: process control and metallurgical modification. Process control approaches can be used to tailor weld pool thermal conditions. These include optimizing the heat input [21], adjusting welding parameters [22,23], and applying external physical fields such as ultrasonic vibration [24] and electromagnetic stirring [25]. Despite their efficacy, these methods demand tight equipment tolerances and stable operation, and their narrow processing windows constrain industrial scalability. In the field of metallurgical modification, relevant studies have predominantly focused on developing filler wires whose chemical compositions differ from that of the base material (BM) to control the microstructure of the weld zone (WZ). Common examples include Al-Si filler wires (e.g., 4043), which enhance interdendritic feeding via Al-Si eutectic formation [26,27], and additions of Sc and Zr to promote heterogeneous nucleation [28,29]. While such approaches partially suppress hot cracking, the resulting chemical heterogeneity in the WZ causes property mismatch with the BM [30,31]. It also increases the risk of local galvanic corrosion due to potential differences [32]. The high cost of strategic elements such as Sc further limits their widespread use [33]. From a sustainability perspective, introducing dissimilar alloying elements with a filler wire contaminates recycled streams, complicating purification and hindering closed-loop recycling [34,35].

Against this backdrop, a welding strategy—ensuring chemical composition consistency between the filler wire and BM—enables property compatibility and supports sustainable alloy recycling. This approach is consistent with the concept of material elementalization and enhances value within a circular economy [36]. However, the inherent solidification cracking tendency of the 6000-series alloys cannot be completely overcome solely by using filler wires that match the composition of the BM [37].

To enhance the weldability of such composition-matched joints, metallurgical strategies can be bifurcated. One path focuses on refining the filler wire itself through advanced alloy design and thermomechanical processing, aiming to improve its intrinsic properties [38]. A more direct and composition-preserving alternative is the incorporation of exogenous ceramic nanoparticles (e.g., TiC, WC). These particles act primarily as inert, physical grain refiners within the weld pool, a mechanism distinct from alloying and evidenced in other joining processes [39].

Recent studies indicate that nanoparticle reinforcement can address this limitation by refining solidification microstructures in WZs. When 7075 aluminum alloy is welded, the introduction of nanoparticles such as TiC and TiB₂ into the filler wire has been

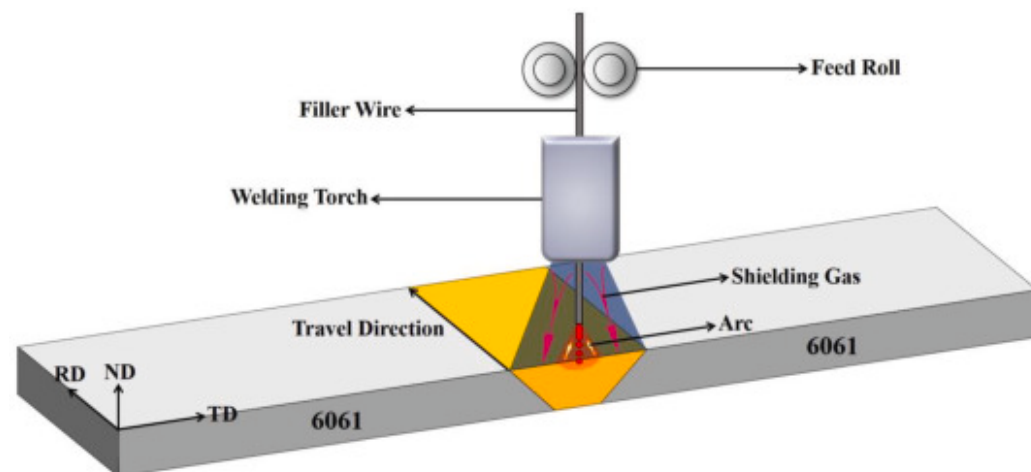
shown to produce significant grain refinement in the WZ [40,41]. For example, the addition of 1.7vol% TiC reduced the average grain size in the WZ from 70.5 μm to 9.4 μm [37]. These particles act as heterogeneous nucleation substrates and promote the formation of equiaxed grains. They also impede grain-boundary migration via Zener pinning, thereby optimizing the distribution of low-melting eutectic phases [41].

Despite these advances, research on the MIG welding process remains limited. A notable gap persists: most studies have targeted 7000-series alloys, whereas the 6000-series is more prevalent industrially and faces acute weldability challenges. Moreover, the influence of nanoparticles on precipitation behavior during post-weld heat treatment (PWHT) is still insufficiently understood. In this work, TiC nanoparticles are incorporated into a 6061-TiC filler wire. This approach systematically reveals the synergistic effects of nanoparticle reinforcement in simultaneously regulating the solidification microstructure and the subsequent precipitation behavior during PWHT in composition-matched joints. The results aim to provide mechanistic insight and experimental evidence supporting high-performance, recyclable welding technologies for 6000-series aluminum alloys.

2. Materials and methods

2.1. Materials and welding procedure

The 6-mm-thick 6061-T6 aluminum alloy plates were joined by MIG welding (Fig. 1). Two types of filler wires with a diameter of 1.2mm were used: a new 6061-TiC filler wire containing 1.5wt% TiC nanoparticles ($\approx 50\text{nm}$) and a mismatched 4043 filler wire. In this study, the joints welded with the matched 6061-TiC filler wire are referred to as similar-metal welded joints (SMW), because the filler metal composition is nominally the same as that of the 6061 BM. The joints welded with mismatched 4043 filler wires are referred to as dissimilar-metal welded joints (DMW), although the BMs on both sides are 6061-T6 aluminum alloys. Here, the word “dissimilar” emphasizes the difference in composition between the filler wire and the BM. The chemical compositions of the BM and filler wires are listed in Table 1. The Ti content (0.42wt%) reported in Table 1 refers specifically to the soluble Ti content in the $\alpha\text{-Al}$ matrix. This value excludes the Ti tied up in undissolved TiC nanoparticles. In this study, a set of parameters corresponding to the industry-standard 4043 filler wire was applied to all joints, in order to directly compare the effects of filler wire composition. These parameters consisted of a welding speed of 50cm/min, a current of 160–170 A, an arc voltage of 22–24V, and high-purity argon (99.999%) as the shielding gas.



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Fig. 1. Schematic of the welding procedure.

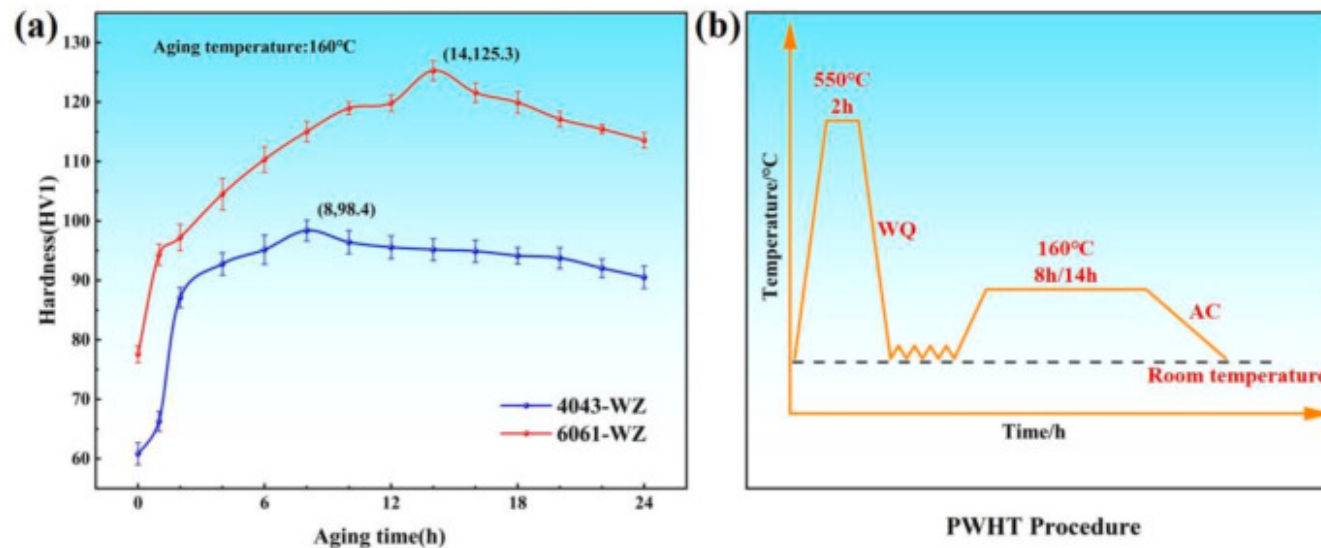
Table 1. Chemical compositions (wt%) of the BM (6061-T6 aluminum alloy) and filler wires (4043 and 6061-TiC).

Alloy	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Al
6061 (BM)	0.60	0.52	0.24	0.12	0.98	0.11	0.13	0.04	Bal.
4043 (filler wire)	5.2	0.13	0.10	0.06	0.02	–	0.15	0.01	Bal.
6061-TiC (filler wire)	0.67	0.12	0.29	0.01	0.98	0.19	0.07	0.42	Bal.

2.2. PWHT procedure

After welding, the joints were subjected to PWHT to evaluate the synergistic effects of TiC nanoparticles and heat treatment on the microstructure and mechanical properties. The PWHT system includes both solid solution treatment and artificial aging treatment. They were first solutionized at 550°C for 2h and then rapidly quenched in water. Artificial aging was subsequently performed at 160°C. To determine the peak aging condition for the WZs of each joint, age-hardening curves (Vickers hardness

versus aging time) were measured at 160°C for regions near the weld centerline. The peak aging time was defined as the aging time corresponding to the maximum average hardness value. The hardness testing method is described in Section 2.3. As shown in Fig. 2a, the maximum average hardness for the 4043-WZ was achieved at 8h, while that for the 6061-WZ was achieved at 14h. Consequently, 8h and 14h were selected as the peak aging times for the 4043 and 6061 joints, respectively. The joints without PWHT were designated as 4043 and 6061 joints, whereas those with PWHT were designated as P-4043 and P-6061 joints, respectively.



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Fig. 2. (a) Age-hardening curves (hardness vs. aging time) for the 4043-WZ and 6061-WZ measured at 160°C, used to determine the peak aging times (8h and 14h, respectively); (b) Schematic of the PWHT procedure: solution treatment at 550°C for 2h, water quenching (WQ), followed by artificial aging at 160°C for various durations, air cooling (AC).

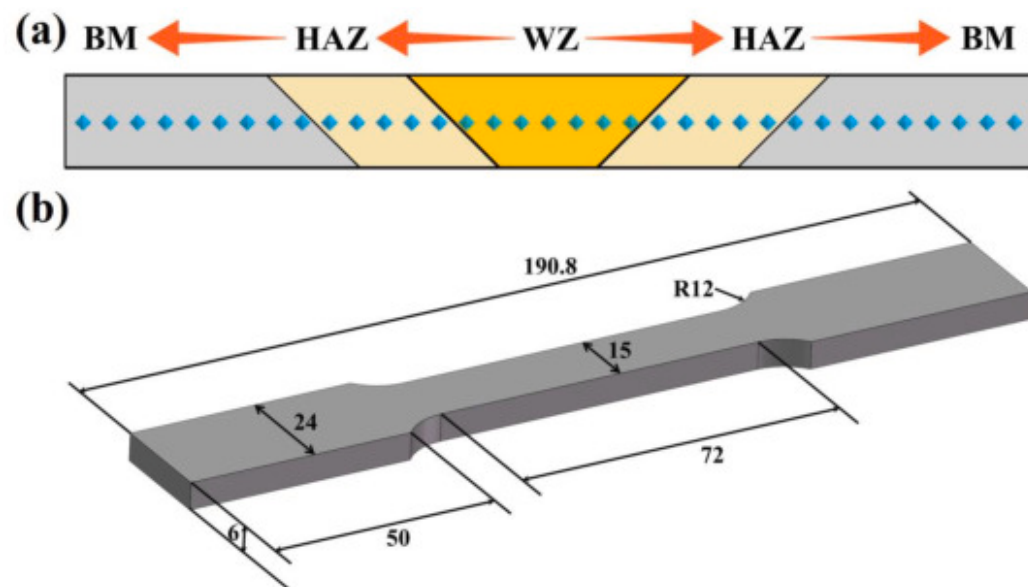
2.3. Microstructural characterization

Cross-sectional specimens for microstructural analysis were extracted from the WZs and heat-affected zones (HAZs) using wire electrical discharge machining (WEDM). The sample surfaces were subjected to a standard metallographic preparation procedure involving grinding, mechanical polishing, and etching in Keller's reagent for 45s. Microstructural observations were carried out

using an optical microscope (OM). Microstructural and fracture surface analyses were performed using a scanning electron microscope (SEM). Electron backscatter diffraction (EBSD) analysis was employed to characterize grain sizes and crystallographic orientation. EBSD measurements were acquired using a Zeiss Sigma field-emission SEM equipped with an Oxford HKL Channel 5 system. Following standard grinding and mechanical polishing, the EBSD samples were electropolished at 20V for 10–15 s in a solution of 10vol% HClO_4 in anhydrous ethanol. The distribution of secondary phases and fine-scale microstructural features in the WZ were examined using a field-emission transmission electron microscope (TEM) operated at 200kV. TEM specimens were prepared from thin slices obtained via WEDM, followed by mechanical grinding to a thickness $<100\mu\text{m}$ and final ion-thinning using a Gatan 695 precision ion-polishing system.

2.4. Mechanical property testing

Vickers hardness was measured using a 200 HVS-5 tester. A 1000g load (HV1) and a 15s dwell time were applied. The hardness values were obtained as the average of at least seven independent indentations made on the polished surface at each location. The specific locations for hardness measurements are illustrated in [Fig. 3](#). Tensile specimens were machined by WEDM to the dimensions specified by the GB/T 228.1-2021 standard ([Fig. 3](#)). The ultimate tensile strength (UTS) was measured using an universal tensile testing machine operating at a crosshead speed of 1 mm/min. Fracture morphology of the tensile specimens was also examined by SEM after testing. Three replicate tests were performed for each welding condition to ensure statistical reliability.



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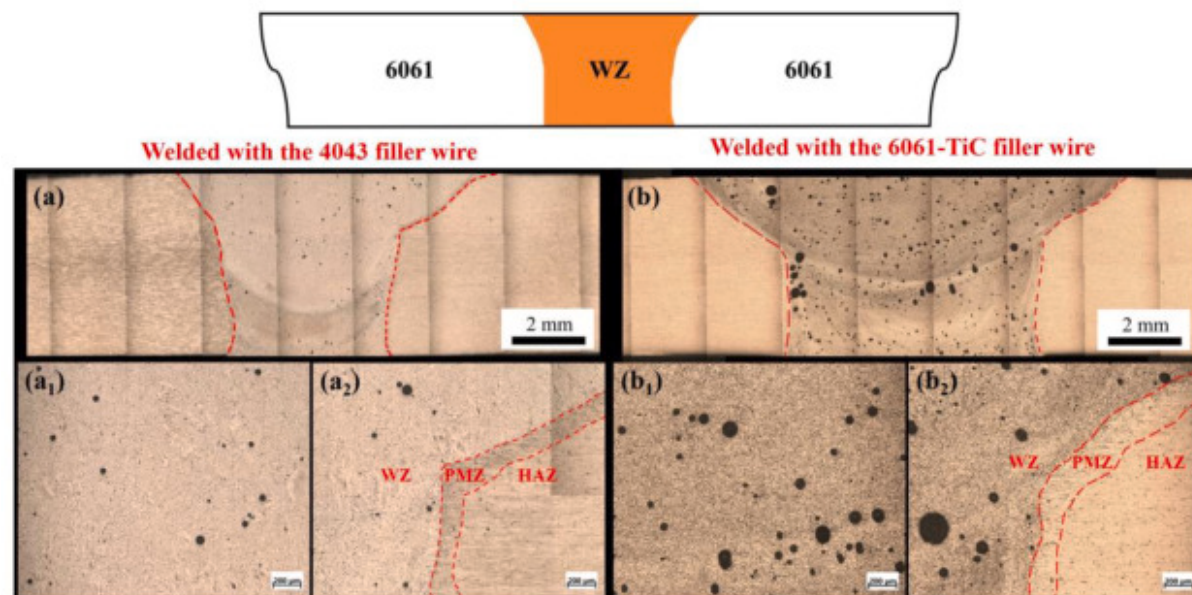
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Fig. 3. (a) Schematic of hardness measurement locations; (b) dimensions of the tensile specimens (mm).

3. Results

3.1. Microstructure

The microstructural characteristics of the joints produced using different filler wires are presented in Fig. 4. The fusion lines (marked by red dashed lines) are symmetric across all joints. The WZ produced with the 4043 filler wire exhibited a characteristic dendritic solidification structure, with coarse, elongated columnar grains. By contrast, the WZ welded with the 6061-TiC filler wire exhibited a significantly refined microstructure consisting of fine, nearly spherical equiaxed grains. The observed microstructural transition suggests that the TiC nanoparticles likely served as potent heterogeneous nucleation sites. Their presence contributed to suppressing dendritic growth and promoting the formation of refined equiaxed grains during solidification.



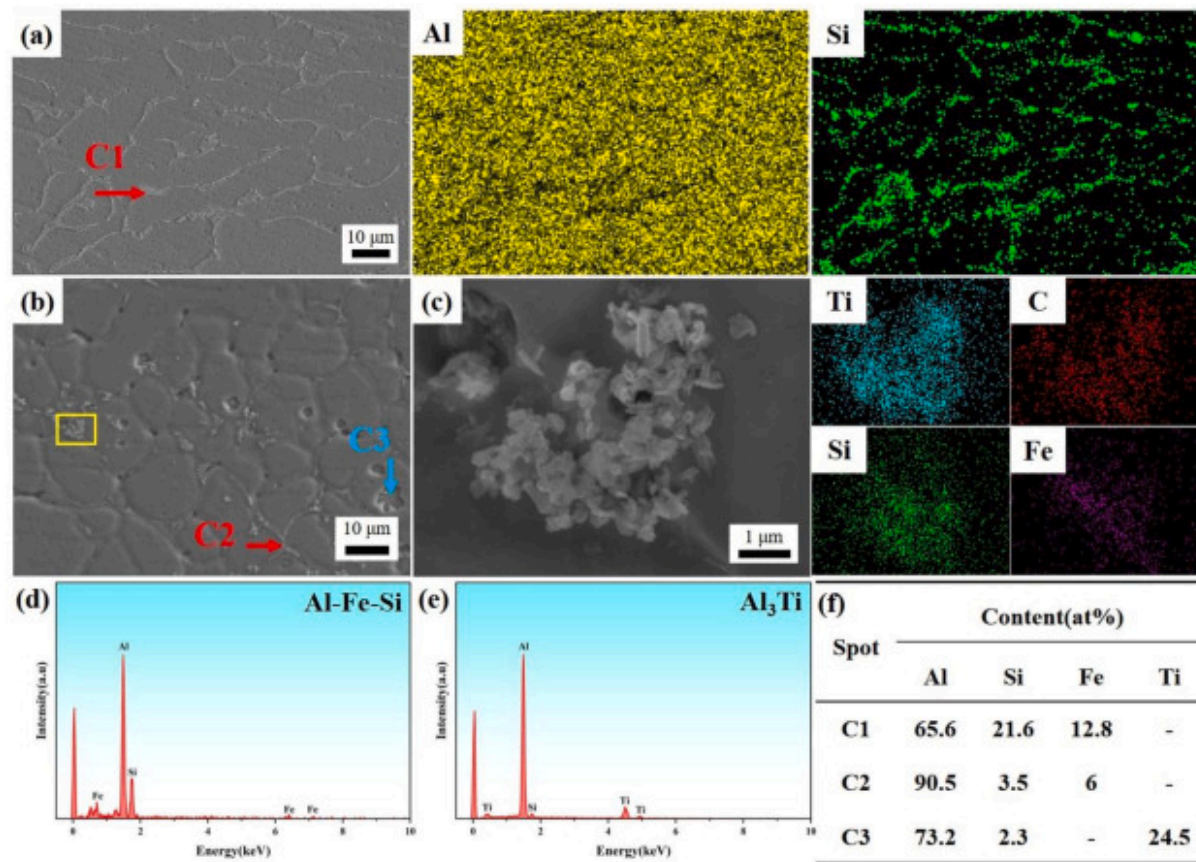
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Fig. 4. Optical micrographs of two types of joints: (a) 4043 joint; (b) 6061 joint.

The profound influence of filler wire composition on the formation of intermetallic compounds in the WZ was investigated using SEM and energy-dispersive X-ray spectroscopy (EDS) (Fig. 5). Red arrows indicate the Al-Fe-Si phase, whereas blue arrows denote the Al_3Ti phase. In the 4043-WZ, a continuous network of the Al-Si eutectic formed along the grain boundaries. Furthermore, blocky particles at grain boundaries were identified by EDS as the Al-Fe-Si phase. This intermetallic phase, together with the eutectic, is thus a characteristic microstructural constituent of the 4043-WZ. By contrast, the 6061-WZ contained Ti-enriched blocky phases; EDS analysis identified the precipitate at point C3 as Al_3Ti and the one at C2 as Al-Fe-Si phase. Furthermore, most of the TiC nanoparticles were located at the grain boundaries, forming bright agglomerates, while a smaller fraction was dispersed within the grains. Elemental mapping (Fig. 5c) revealed enrichment of Ti and C in the grain-boundary particles, with minor amounts of Si and Fe. This confirms that these particles are TiC nanoparticle clusters ($\approx 300\text{--}500\text{nm}$) that coexist with other intermetallic compounds. Consistent with previous studies [42,43], TiC nanoparticles segregate to the grain boundaries, forming interconnected networks (Fig. 5b). Particle-pushing theory accounts for this spatial arrangement. As the solid-liquid

interface advances, it rejects nanoparticles, which then accumulate at the grain boundaries during solidification. Grain boundaries are rich in defects and thus serve as effective nucleation sites for secondary phases. This leads to the observed spatial correlation between nanoparticles and grain-boundary precipitates.



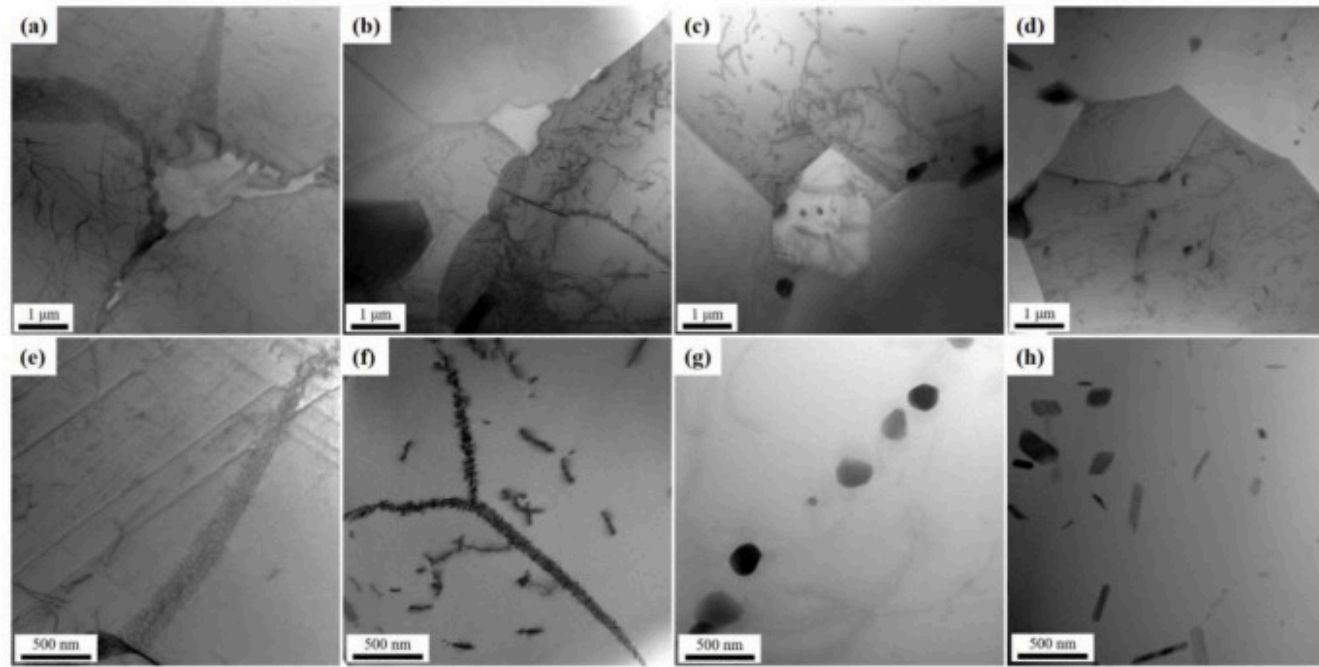
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Fig. 5. SEM-EDS of secondary phases in WZs: (a) 4043-WZ showing Al-Si eutectic and Al-Fe-Si phase (red arrows); (b) 6061-WZ showing Al₃Ti (blue arrows) and TiC clusters; (c) magnified region of (b) with elemental maps; (d-e) point spectra confirming Al-Fe-Si and Al₃Ti; (f) quantitative compositions (at%) at C1-C3. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

To clarify the actual contribution of TiC nanoparticles to the heterogeneous nucleation of α -Al, the effective volume fraction of well-dispersed TiC (f_{eff}) was quantitatively evaluated through compositional analysis and image statistics. The analysis revealed that, despite a nominal addition of 1.5wt% (0.83vol%) TiC, the actual retained TiC volume fraction was only 0.56vol% due to partial dissolution of Ti into the matrix. Further analysis indicated the presence of TiC agglomeration, with well-dispersed particles accounting for only 38.2% of the total area fraction. Based on this, the calculated f_{eff} available as potent nucleation sites was merely 0.215vol%, which represents only 26% of the nominal addition. The agglomeration weakened the heterogeneous nucleation efficiency of TiC. This is primarily because agglomeration not only significantly reduces the number of independent nucleation sites in the system, but also shields the interfaces of particles within clusters, preventing them from providing sufficient heterogeneous nucleation surfaces for the melt. Consequently, despite the inherent grain refinement potential of TiC, its agglomerating behavior limits the practical effectiveness in refining α -Al grains.

The evolution of precipitates and dislocation structures in the WZ before and after PWHT was further examined by TEM, as shown in Fig. 6. Rapid cooling during welding produced a typical cast microstructure in the as-welded 4043-WZ. Coarse secondary phases formed along the grain boundaries (Fig. 6a), thereby degrading the mechanical properties. The welding thermal cycle produced rapid cooling, accompanied by thermal and solidification shrinkage stresses [[44], [45], [46]], resulting in a high dislocation density. These dislocations organized into extensive dislocation walls and subgrain boundaries (Fig. 6e). By contrast, the 6061-WZ contained a higher density of fine, uniformly distributed precipitates, facilitated by the presence of TiC nanoparticles. It also displayed a more complex dislocation substructure, consisting of entangled dislocations and stable subgrain networks (Fig. 6f).



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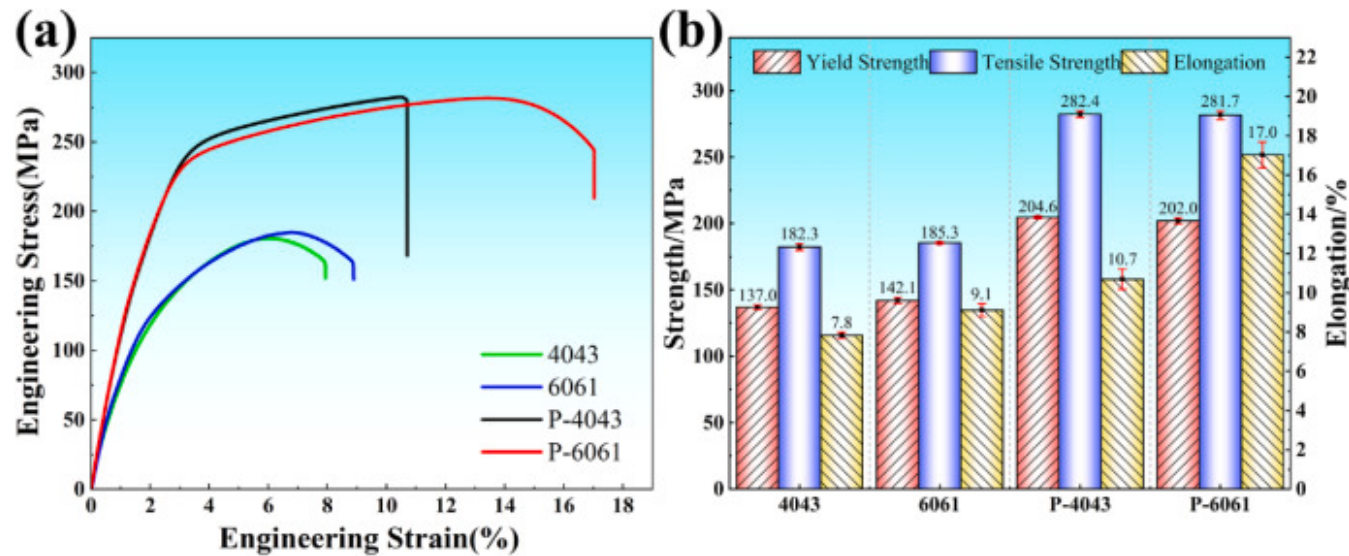
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Fig. 6. TEM micrographs of WZs: (a, e) 4043-WZ; (b, f) 6061-WZ; (c, g) P-4043-WZ; (d, h) P-6061-WZ.

Following PWHT, the grain-boundary precipitates in the P-4043-WZ underwent coarsening. Concurrently, dislocation recovery substantially reduced dislocation density, thereby yielding a more uniform dislocation distribution. However, the Al-Si eutectic phases retained the ability to pin dislocations at grain boundaries. By contrast, the P-6061-WZ underwent a pronounced microstructural transformation (Fig. 6d and h). It exhibited fine, continuous grain-boundary precipitates and a high density of uniformly dispersed intragranular precipitates. These intragranular precipitates were comparatively finer and exhibited rod-like or irregular, disk-shaped morphologies. The refined, uniform distribution of precipitates indicates a more homogeneous microstructure in the P-6061-WZ. This effectively mitigates the degradation of mechanical properties typically caused by elemental segregation.

3.2. Mechanical properties

The macroscopic mechanical properties were evaluated by tensile testing (Fig. 7). As shown in Fig. 7b, the yield strength (142.1 ± 2.3 MPa) and tensile strength (185.3 ± 0.6 MPa) of the 6061 joint were both slightly higher than those of the 4043 joint. After PWHT, the tensile strength values of the P-6061 and P-4043 joints were very similar, at 281.7 ± 3.2 MPa and 282.4 ± 2.5 MPa, respectively. The P-6061 joint exhibited significantly higher elongation ($17.0 \pm 0.7\%$) than the P-4043 joint ($10.7 \pm 0.5\%$), thereby indicating improved ductility of the former.

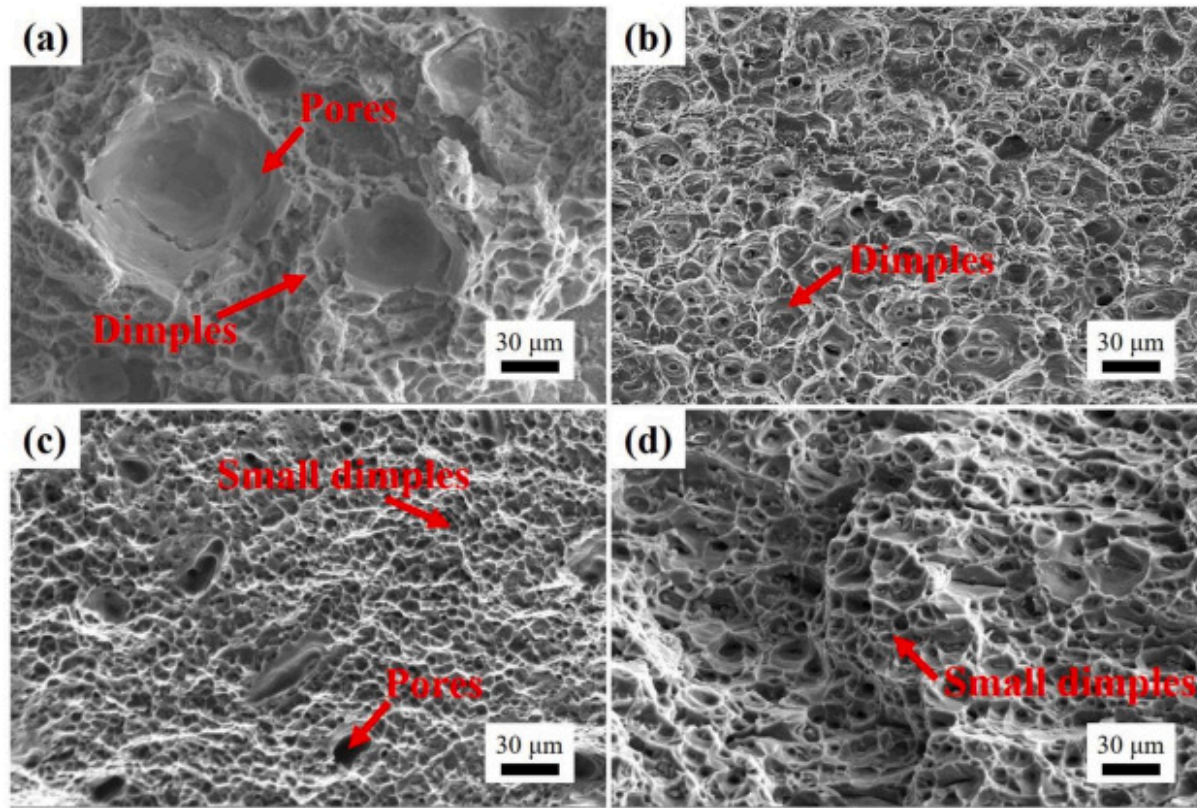


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Fig. 7. (a) Engineering stress–strain curves for the joints; (b) summary of mechanical properties.

The fracture morphologies of the joints are presented in Fig. 8. All joints failed through a predominantly ductile fracture mechanism, as characterized by dimples formed by microvoid coalescence, though with distinct morphological variations. The 4043 joint exhibits a fracture morphology dominated by pores, which is consistent with its limited ductility. By contrast, the 6061 joint revealed a high density of fine, equiaxed dimples, indicative of a superior ductile fracture mode and the enhanced ductility measured in the tensile tests. After PWHT, the P-4043 joint exhibited finer dimples, but its ductility was limited by numerous pores (Fig. 8c). By contrast, the P-6061 joint exhibited uniformly distributed, fine, deep dimples (Fig. 8d). This morphology provides clear evidence of extensive, homogeneous plastic deformation prior to fracture, consistent with the superior tensile elongation.



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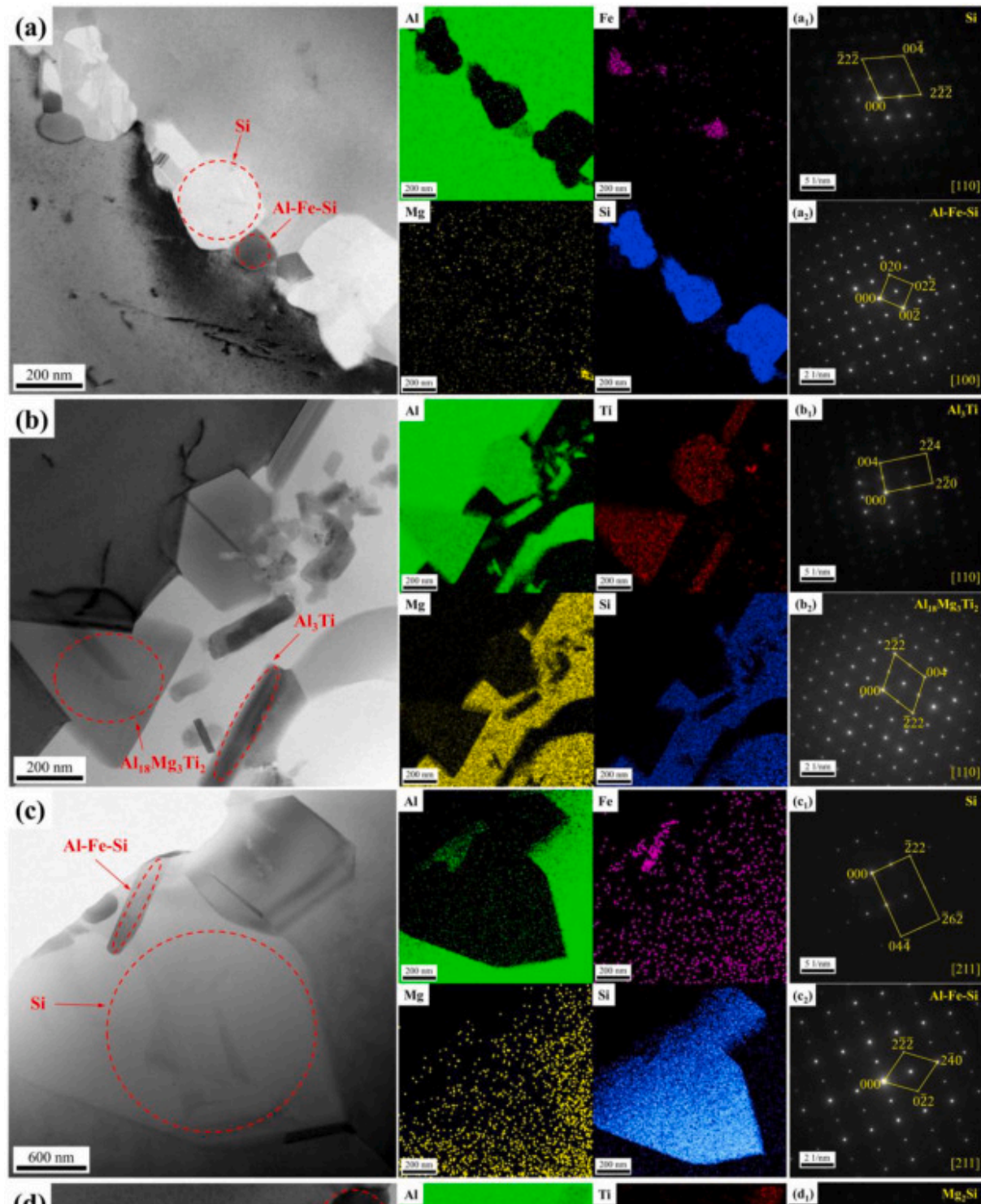
Fig. 8. Tensile fracture morphologies of the joints: (a) 4043 joint; (b) 6061 joint; (c) P-4043 joint; (d) P-6061 joint.

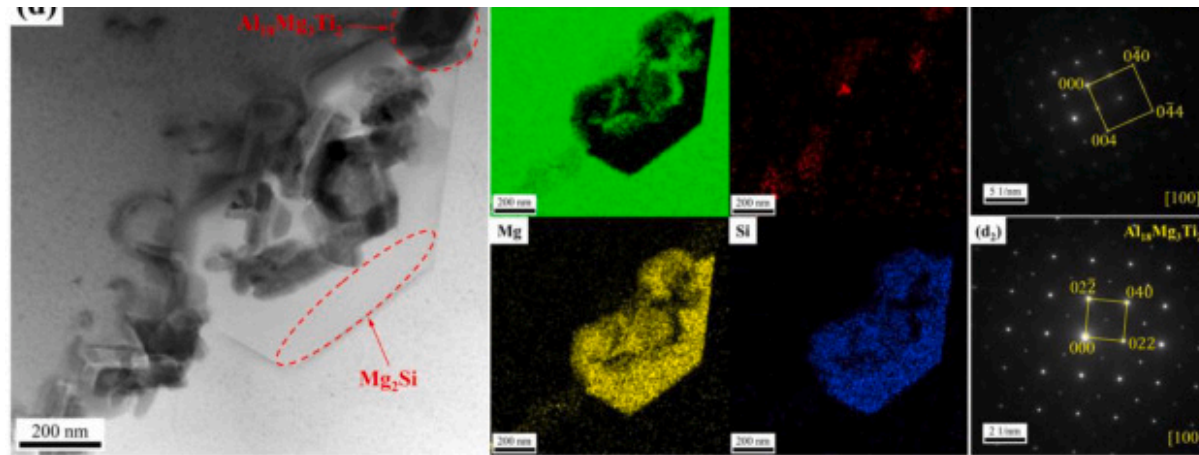
4. Discussion

4.1. Evolution of precipitates

The characteristics of secondary phases in the WZs were examined to correlate microstructural evolution with mechanical properties. Fig. 9 presents representative high-magnification TEM images focusing on the grain-boundary regions in the WZs,

both in the as-welded condition and after PWHT. The types and distributions of grain-boundary precipitates were determined using selected-area electron diffraction (SAED).





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Fig. 9. TEM micrographs of precipitates in the WZs: (a) 4043-WZ; (b) 6061-WZ; (c) P-4043-WZ; (d) P-6061-WZ.

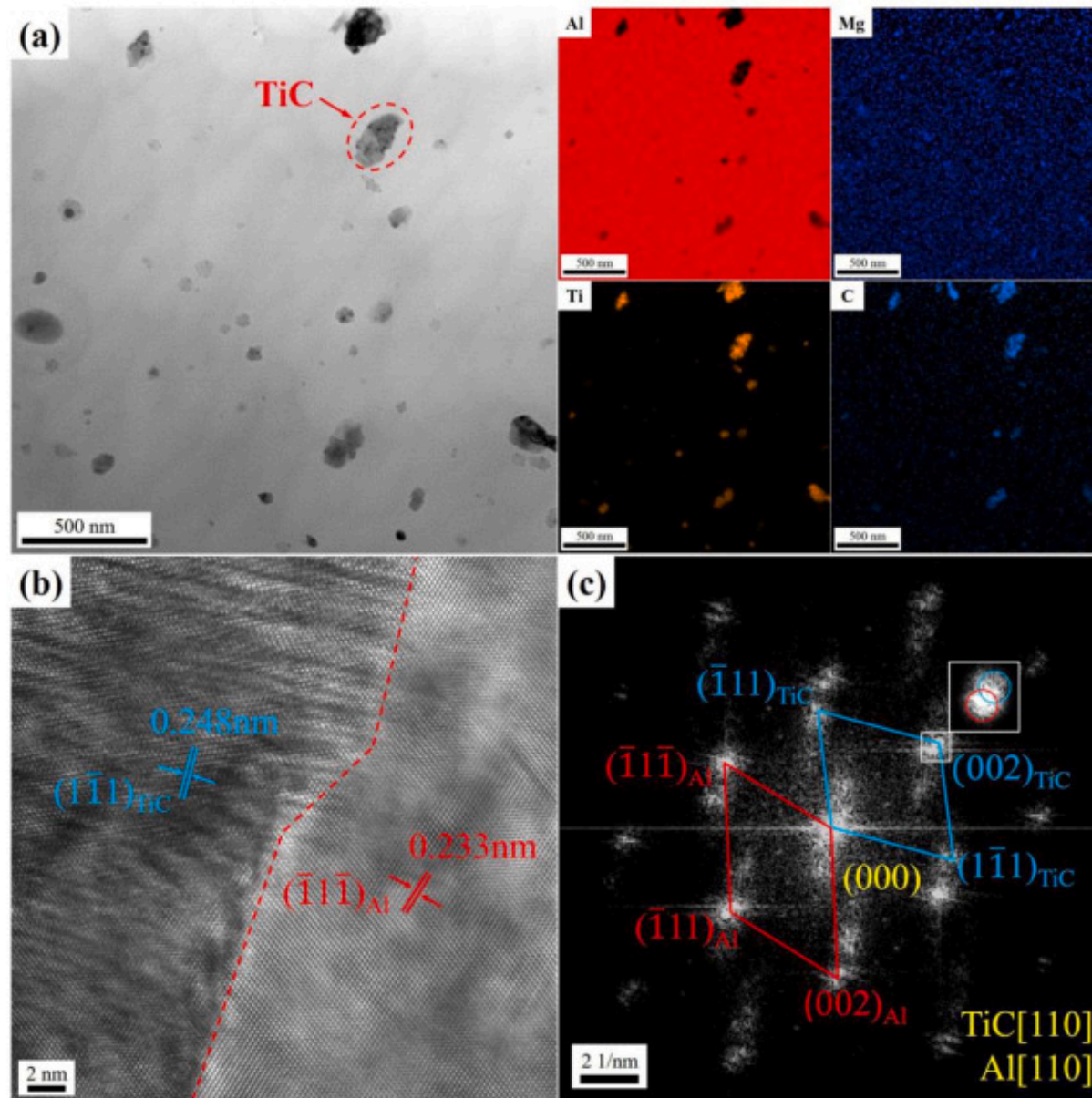
Fig. 9a reveals two distinct types of precipitates along the grain boundaries in the 4043-WZ. These were identified via SAED as a Si-rich Al-Si eutectic and the Al-Fe-Si phase. The low Mg and high Si contents in the 4043-WZ suppress the precipitation of Mg_2Si . By contrast, the 6061-WZ contained well-dispersed TiC nanoparticles throughout the matrix, constituting a thermodynamically stable reinforcing phase. Fig. 9b displays blocky precipitates enriched in Ti and Mg. The corresponding SAED pattern in Fig. 9b₂ identifies this phase as $\text{Al}_{18}\text{Mg}_3\text{Ti}_2$. According to previous reports [42,47,48], $\text{Al}_{18}\text{Mg}_3\text{Ti}_2$ serves as an effective

heterogeneous nucleation site for other precipitates. It also segregates to grain boundaries, where it pins them and retards grain growth. TEM analysis further revealed that the Al_3Ti phase in the 6061-WZ possesses a tetragonal D0_{22} crystal structure. Nanometric $\text{D0}_{22}\text{-Al}_3\text{Ti}$ particles are known to act as potent heterogeneous nucleation sites for the $\alpha\text{-Al}$ matrix, thereby facilitating significant grain refinement [49].

During the PWHT of the P-4043-WZ, a partial dissolution of the Al-Si eutectic into the Al matrix occurred. However, owing to the limited solid solubility of Si in Al [50], most of the Si remained undissolved and underwent coarsening to reduce the overall interfacial energy. This coarsening proceeds via Ostwald ripening [[51], [52], [53]]. Smaller precipitates dissolve, releasing solute atoms that diffuse to and promote the growth of larger precipitates. Finally, the dominant larger precipitate phases are the Al-Si eutectic and Al-Fe-Si phase.

In the P-6061-WZ, SAED identified blocky, nanoscale grain-boundary precipitates as Mg_2Si (Fig. 9d₁). During PWHT, the $\text{Al}_{18}\text{Mg}_3\text{Ti}_2$ and Al_3Ti phase promoted the precipitation of finer, more densely distributed Mg_2Si strengthening precipitates.

To provide deeper characterization of the TiC nanoparticles and their interface with the $\alpha\text{-Al}$ matrix, high-resolution transmission electron microscopy (HRTEM) and nano-EDS analyses were performed on the 6061-WZ. Enrichment of Ti and C elements was observed at the particulate site in Fig. 10a. Combined with its HRTEM image (Fig. 10b) and Fast Fourier Transform (FFT) analysis (Fig. 10c), this particulate was identified as TiC. The interface between TiC and $\alpha\text{-Al}$ was clean and intimately bonded. Crystallographic analysis indicated an orientation relationship between the TiC nanoparticle and the $\alpha\text{-Al}$ matrix satisfying $[110]\text{TiC} // [110]\text{Al}$. The measured interplanar spacings for $(1\ \bar{1}\ 1)\text{TiC}$ and $(\bar{1}\ 1\ \bar{1})\text{Al}$ were 0.248 nm and 0.233 nm, respectively.



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Fig. 10. (a) TEM image and corresponding EDS elemental maps of a TiC nanoparticle; (b) HRTEM image of the interface between TiC and the α -Al matrix; (c) the corresponding FFT pattern from the region in (b).

Since both TiC and α -Al possess an FCC crystal structure, their interfacial mismatch was calculated using the Bramfitt two-dimensional mismatch model. The calculation formula is [43]:

$$\delta = \frac{(d_{\text{TiC}} - d_{\text{Al}})}{d_{\text{TiC}}} \times 100\% \quad (1)$$

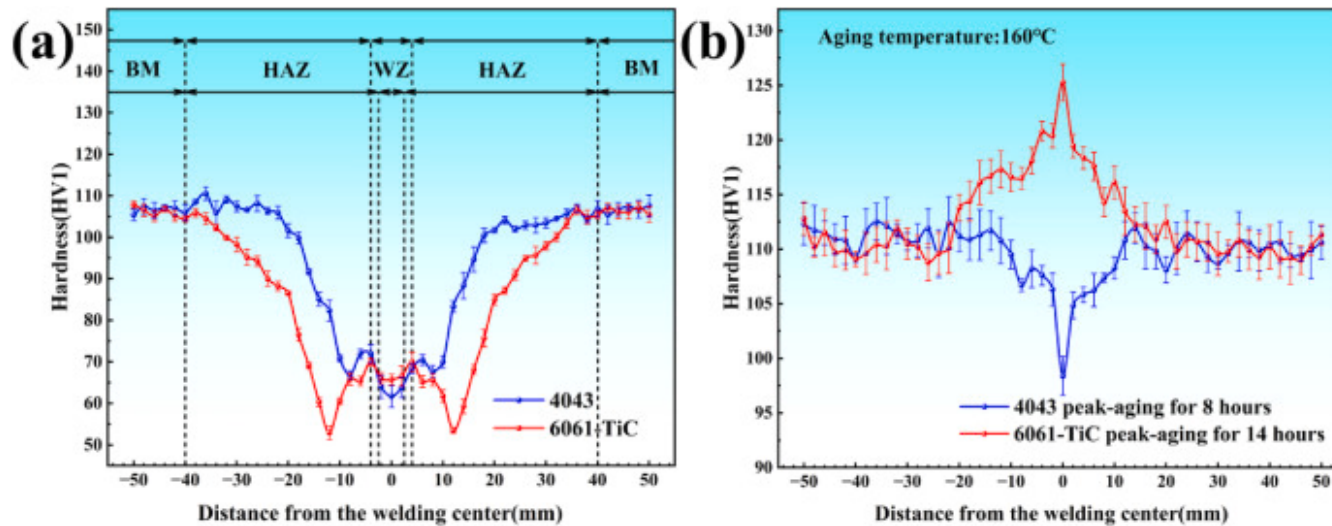
where δ is the interfacial mismatch, and d_{TiC} and d_{Al} are the measured interplanar spacings of TiC and α -Al mentioned above. The calculated δ was 6.05%, which falls within the typical range (6%–12%) for forming a semi-coherent interface [42,54,55]. This suggests the formation of a low-misfit semi-coherent interface between TiC and α -Al. This interfacial structure confers a dual benefit: on one hand, it can effectively serve as a heterogeneous nucleation site for α -Al grains, promoting grain refinement of the solidified microstructure [43]; on the other hand, its sound interfacial bonding ensures interfacial stability during loading, facilitating efficient load transfer from the matrix to the reinforcement.

4.2. Mechanistic analysis of strength and ductility

Building on the microstructural evolution described in Fig. 16, this section correlates these changes with the resulting mechanical properties.

Before and after PWHT, the tensile specimens of the 6061 joints fractured in the HAZ, whereas those of the 4043 joints fractured in the WZ. After PWHT, the strength of both joints was significantly improved, and the elongation of the P-6061 joint and the P-4043 joint was 17.0 % and 10.7 %, respectively, indicating a pronounced difference in the overall mechanical properties of the two joints. To evaluate the effect of PWHT on the strength and plastic deformation behavior of the joints, the hardness distribution across the BM-HAZ-WZ region was measured to characterize the local strength level and its spatial distribution. EBSD analysis, including grain size statistics, grain boundary type, and texture orientation characterization, was further employed to quantify the degree of microstructural refinement and uniformity. Based on these parameters, together with the tensile properties and fracture locations, the effect of PWHT and filler wire selection on the strength and plasticity of the different joints can be systematically evaluated.

The influence of filler wire composition and PWHT on mechanical properties was evaluated using Vickers hardness profiling (Fig. 11). Both as-welded joints exhibited a characteristic W-shaped hardness profile (Fig. 11a). The hardness profile increased, decreased, and then increased again, ultimately stabilizing at approximately 107 HV1, comparable to the BM. A pronounced hardness drop occurred in the WZ and HAZ, which is primarily attributed to thermal softening induced by the welding thermal cycle. The hardness of the 6061-WZ exceeded that of the 4043-WZ by approximately 5 HV1. After PWHT, the P-6061-WZ exhibited significantly higher hardness than the P-4043-WZ (Fig. 11b). This reflects the strong synergy between TiC nanoparticles and the aging treatment.



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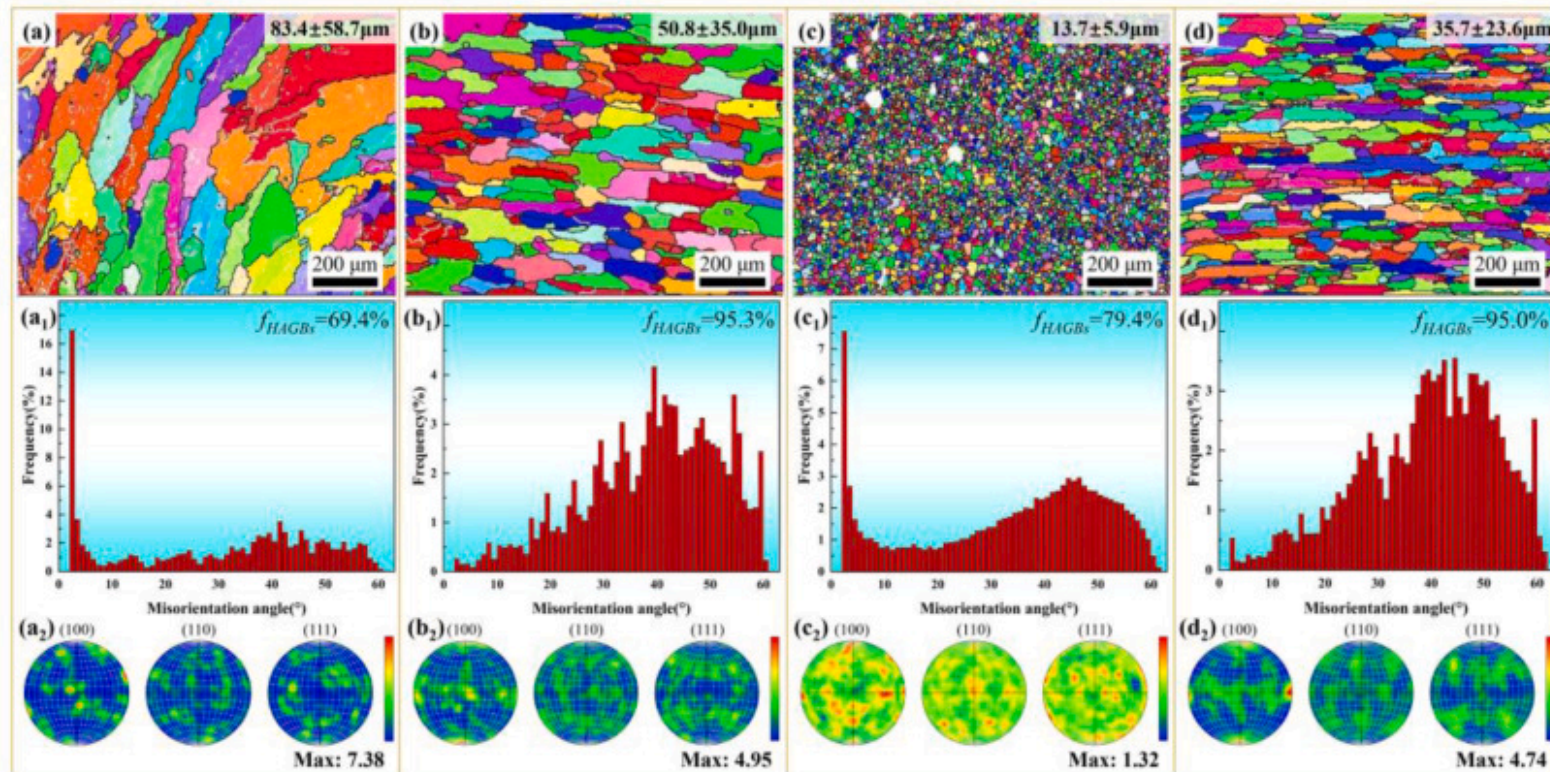
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Fig. 11. Vickers hardness profiles of joints: (a) as-welded; (b) after PWHT.

From the hardness distribution of the joints, it can be seen that although numerous pores are present in the WZ, the lowest hardness of the 6061 joints occurs in the HAZ (53 HV1) rather than in the WZ (62 HV1). This trend becomes more pronounced after PWHT, and the hardness of the WZ (125 HV1) is significantly higher than that of the HAZ (110 HV1). This indicates that the HAZ becomes the weakest region, which correlates well with failure in the HAZ during the tensile test. By contrast, the 4043 joint exhibits the lowest hardness in the WZ, which is consistent with the fracture location observed in this joint. At the same time, the average hardness of 4043-WZ after PWHT increased from approximately 62 HV1 to approximately 98 HV1, indicating a

significant enhancement in the deformation resistance of the WZ. The average hardness of the HAZ increased from 68 HV1 to 110 HV1, and the extent of softening was significantly reduced. For the 4043 joint, local necking first occurs in the WZ with the lowest hardness, where cracks first initiate, resulting in low joint plasticity. The 6061 joint exhibits the highest hardness in the WZ. Although the HAZ is relatively weak, it can still accommodate greater deformation, and the plasticity of the joint is significantly higher than that of the 4043 joint.

EBSD grain size analysis shows that the WZ of 6061 joint is significantly refined compared with HAZ (Fig. 12). Fine equiaxed grains imply a large number of grain boundaries, indicating that the grain slip paths are numerous, deformation compatibility is improved, and plasticity is enhanced [[56], [57], [58]]. In the 6061-WZ, the TiC nanoparticles are attributed with grain refinement, most likely through their role as heterogeneous nucleation sites for α -Al and via the pinning of grain boundaries and hindrance of dislocation motion [59,60].

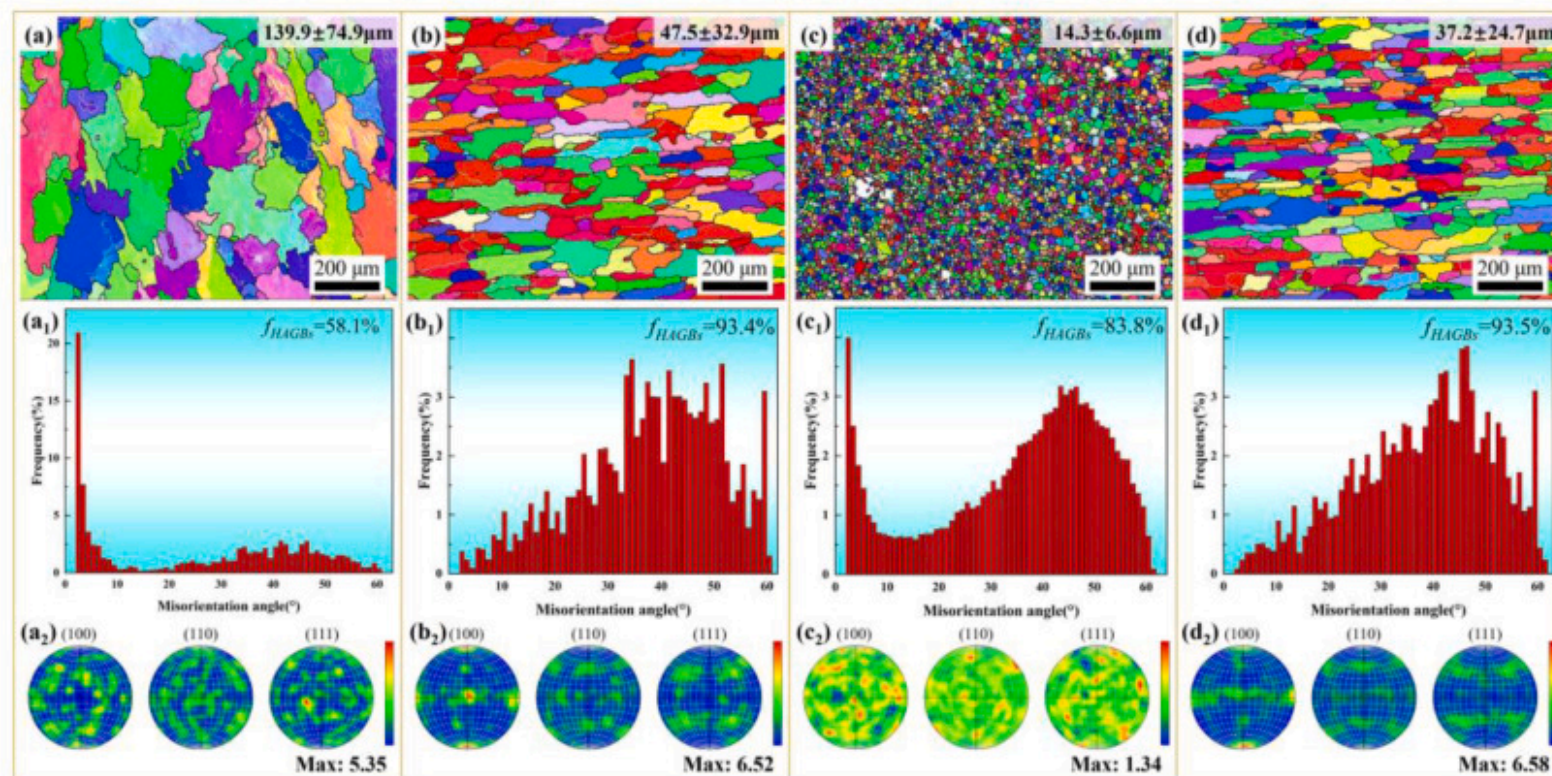


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Fig. 12. EBSD analysis of the microstructure and crystallographic characteristics in different regions in the as-welded condition: (a) 4043-WZ; (b) 4043-HAZ; (c) 6061-WZ; (d) 6061-HAZ.

This grain refinement in the WZ leads to a simultaneous increase in strength and plasticity via the Hall-Petch effect, which is consistent with fracture occurring in the HAZ. For the 4043 joint, the grain size in the WZ is significantly larger than that in the HAZ, resulting in fracture of the joint in the WZ. After PWHT, the grain size of the material did not undergo any significant change (Fig. 13). On the other hand, the high-angle grain boundaries in 6061-WZ can effectively hinder crack propagation and promote crack deflection and branching. Thus, the crack propagation path is significantly extended, and additional fracture work is consumed, thereby ultimately improving the plasticity of the material [61,62]. This is also illustrated by the microporous aggregation fracture observed on the tensile fracture surface. Therefore, grain refinement in 6061-WZ can enhance the tensile properties of the joints by increasing resistance to crack initiation and propagation and simultaneously improving their strength and plasticity.



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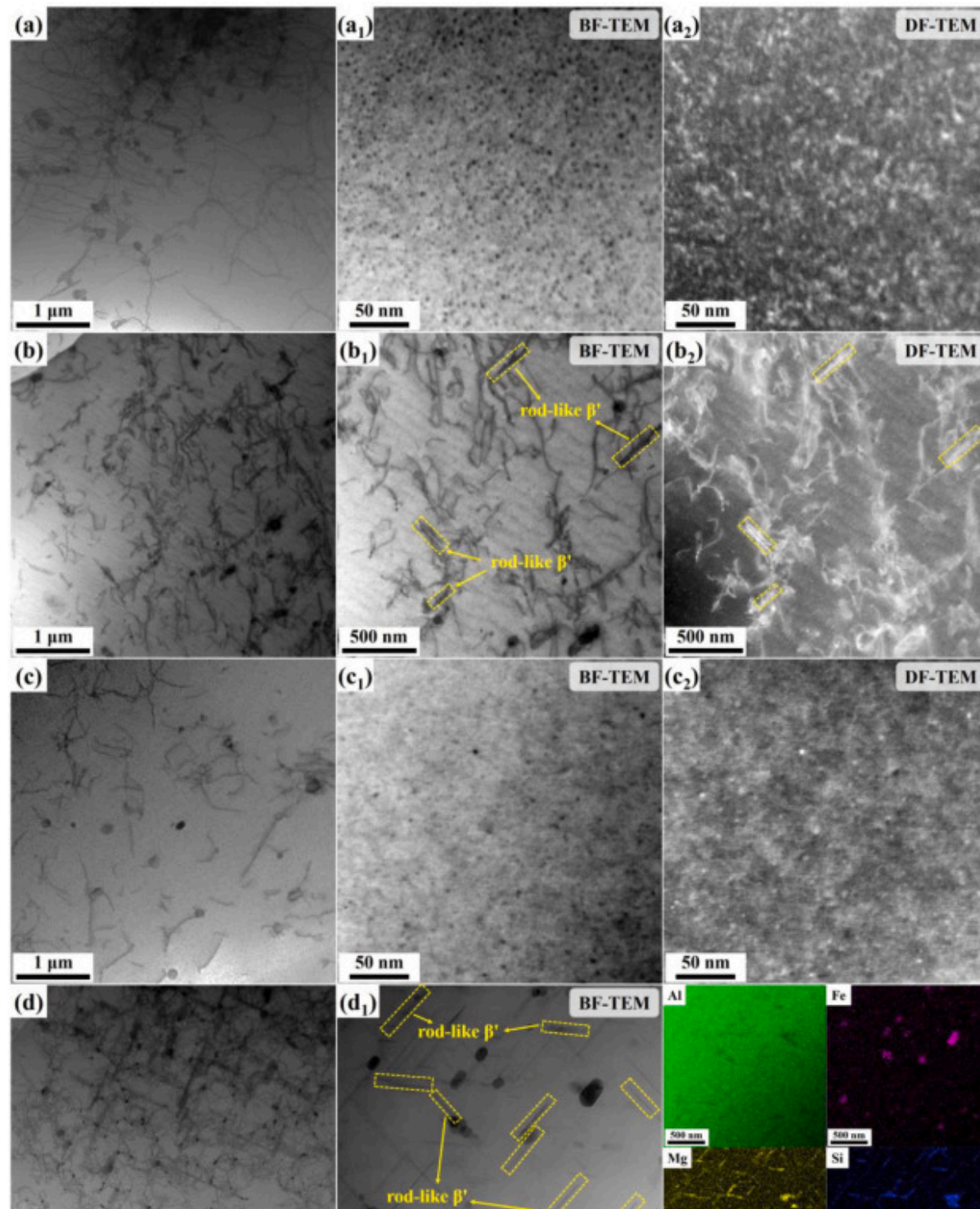
Fig. 13. EBSD analysis of the microstructure and crystallographic characteristics in different regions under the PWHT condition: (a) P-4043-WZ; (b) P-4043-HAZ; (c) P-6061-WZ; (d) P-6061-HAZ.

EBSD texture analysis shows that for the 6061 joints, the grain orientation distribution in the WZ is approximately random, with an extremely weak texture, and the maximum pole density is only 1.3 mud. A strong fiber texture (4.7 mud) aligned with the applied loading direction is formed in the HAZ, which is beneficial to slip and may further reduce the macroscopic yield strength of the HAZ. By contrast, 4043-WZ exhibits a pronounced preferred orientation, with a texture strength as high as 7.4 mud and evident aggregation of {100} and {111}. After PWHT, the texture also exhibits a similar trend.

This difference in texture exerts a significant influence on plastic deformation. In 6061-WZ, the near-random texture and fine equiaxed grains allow a large number of differently oriented grains to participate in plastic deformation on multiple slip systems. This is beneficial for the coordinated deformation between adjacent grains and promotes a more uniform distribution of strain [62,63]. Therefore, the WZ can sustain a high overall plastic strain prior to the onset of necking. By contrast, the strong texture in 4043-WZ implies that a large proportion of the grains share similar crystallographic orientations. Under uniaxial tension, plastic deformation is concentrated in specific grain clusters and local banded regions along certain preferred orientations, whereas other grains undergo only relatively small deformation [64,65]. This local strain concentration, coupled with the presence of brittle Al-Si eutectic and coarse Si-rich particles in 4043-WZ, leads to premature crack initiation. This accelerates the failure process of the joint, resulting in the ductility of the 4043 joint being significantly lower than that of the 6061 joint.

From the perspective of fine precipitates, no fine precipitates were found in the as-welded WZ of the 6061 joints produced with 6061-TiC filler wire (Fig. 14). This deficiency is attributed to the rapid cooling of the WZ during the welding process, such that solute atoms such as Mg and Si do not precipitate in time [[66], [67], [68]]. This leads to complete dissolution of the β'' phase in the α -Al matrix, thus preventing the formation of the fine strengthening phase. It is worth noting that the WZ has a chemical composition similar to that of the 6061 BM, and that Al_3Ti and $\text{Al}_{18}\text{Mg}_3\text{Ti}_2$ promotes the precipitation of fine strengthening phases. This enables the precipitation of a fine and relatively uniform distribution of the β'' strengthening phase after PWHT. The β'' phase is fully coherent with the aluminum matrix and is the primary strengthening phase in 6061 aluminum alloy [69,70]. In accordance with classical strengthening theory [[71], [72], [73], [74]], precipitate strengthening increases as particle size decreases. These uniformly distributed nanoscale precipitates enable dislocations to bypass or shear them in a progressive manner, thereby providing an effective obstacle to dislocation motion. This makes an important contribution to precipitation strengthening and enhances the mechanical properties of the joint. At the same time, the introduction of a nano-scale β'' strengthening phase enables the material to maintain a high work-hardening capability while achieving high strength [[75], [76], [77]]. It can accommodate highly uniform plastic deformation prior to necking, thus ensuring excellent uniform plastic deformation capability [78,79]. In the HAZ, grain coarsening occurs during the welding thermal cycle, the precipitated phase coarsens, and the β'' phase is transformed into β' [68,70,80]. At the same time, coarser round and elongated precipitates can be observed in the EDS, which leads to a pronounced softening in the HAZ and significantly reduces the local strength. In the process of PWHT, the coarse precipitated phases are dissolved during the solid-solution treatment, and a dispersed strengthening phase is subsequently obtained during the aging treatment. The strength of the P-6061-HAZ increases; however, it remains lower

than that of the P-6061-WZ. Therefore, the local load-bearing capacity of the HAZ is lower than that of the pore-containing WZ, and fracture preferentially occurs in the HAZ.





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Fig. 14. TEM analysis of the precipitate and EDS in different regions in the as-welded condition: (a) 4043-WZ; (b) 4043-HAZ; (c) 6061-WZ; (d) 6061-HAZ.

By contrast, for the 4043 joint, the WZ is composed of the Al-Si component from the 4043 filler wire and the Al-Mg-Si component from the 6061 BM. As a result, the WZ has a lower Mg content and a higher Si level, which limits the formation of the Mg_2Si strengthening phase and therefore provides only limited precipitation strengthening capability. In the as-welded state, the WZ exhibits a relatively coarse microstructure, characterized by an as-cast structure composed of α -Al dendrites, brittle Al-Si eutectic, and coarser Si-rich particles, with a low hardness of approximately 62 HV1. These coarse, often irregularly shaped second phases act as preferred sites for void nucleation and crack initiation under tensile loading. They do not provide uniform, fine-scale obstacles to dislocation motion, but instead promote local stress and strain concentrations at the particle-matrix interfaces [[81], [82], [83]]. Therefore, the plastic deformation in 4043-WZ tends to localize rapidly around these particles, resulting in early damage and a reduced strain-hardening capacity, thereby limiting the overall plasticity of the joint. During PWHT, the solution treatment dissolves part of the non-equilibrium eutectic structure and increases the contents of Mg and Si in the α -Al matrix. Subsequent artificial aging promotes the precipitation of fine β'' strengthening phase, providing a limited precipitation-strengthening effect. In addition, the uniform distribution of solute atoms and the partial relaxation of welding residual stresses further promote hardening of the WZ [66,84,85]. Therefore, the hardness of the 4043-WZ after PWHT increased from 62 HV1 to 98 HV1, but remained lower than that of P-BM (110 HV1) and P-6061-WZ (125 HV1). The improvement in the strength of the 4043 joint after PWHT is mainly attributed to the recovery of precipitation strengthening in the HAZ and the homogenization and stress relaxation of the WZ, rather than to any significant precipitation-strengthening effect of the 4043-WZ itself. At the same time, the brittle Al-Si eutectic and coarse Si particles in the WZ remained essentially unchanged. Therefore,

although the porosity level in the WZ is low, the intrinsic strength of the WZ remains lower than that of the HAZ. The WZ thus becomes the weakest region of the 4043 joint, and tensile fracture occurs in the WZ.

To quantitatively assess the precipitation behavior of the β'' strengthening phase after PWHT, the area density of precipitates was statistically analyzed in representative regions of the P-6061-WZ and the P-4043-WZ; the results are summarized in Table 2. As shown in Table 2, the area density of β'' precipitates in the P-6061-WZ is 5.1 times that in the P-4043-WZ. Combined with the observations in Fig. 15, the β'' precipitates in the P-6061-WZ exhibit a high-density, uniform, and dispersed distribution. By contrast, the precipitates in the P-4043-WZ are significantly fewer in number, sparsely distributed, and more random. These quantitative data visually reveal the substantial difference in precipitation strengthening response between the two zones.

Table 2. Statistical area density of β'' precipitates in different WZs after PWHT.

Sample	Area Density ($10^2/\mu\text{m}^2$)
P-6061-WZ	17.8 ± 3.2
P-4043-WZ	3.48 ± 0.25

Careful TEM analysis confirms that while Ti-related intermetallics (Al_3Ti and $\text{Al}_{18}\text{Mg}_3\text{Ti}_2$) coexist with nanoscale β'' precipitates in the P-6061-WZ, the β'' phase is generally dispersed throughout the matrix without significant enrichment at the intermetallic/matrix interfaces. This observation strongly suggests that these particles are not the primary heterogeneous nucleation sites for β'' [86]. Instead, their role in facilitating precipitation is indirect, operating through distinct mechanisms that influence the precipitation environment and kinetics. The respective contributions of $\text{Al}_{18}\text{Mg}_3\text{Ti}_2$ and Al_3Ti can be differentiated as follows [87].

The $\text{Al}_{18}\text{Mg}_3\text{Ti}_2$ phase likely influences precipitation through solute partitioning. As a Mg-containing compound, its formation and stability during solidification and PWHT can locally alter the effective concentration and supersaturation of Mg in the surrounding matrix. By acting as a partial sink for Mg, it may modify the conditions for the formation of early-stage Mg/Si solute clusters, thereby affecting the nucleation rate and spatial distribution of β'' [88].

By contrast, the Al_3Ti phase primarily contributes through microstructural refinement. Its well-known role as a potent grain refiner during weld pool solidification leads to a significantly increased grain boundary density in the WZ. These grain boundaries serve as fast diffusion pathways for solute atoms (Mg, Si) and vacancies during the subsequent PWHT. This enhanced diffusion kinetics promotes a more uniform and widespread nucleation of β'' precipitates across the refined grains, facilitating their high-density, dispersed distribution.

In summary, the promotion of high-density β'' precipitation is not a result of direct heterogeneous nucleation on these Ti-rich phases. It arises from their synergistic, yet distinct, indirect roles: $\text{Al}_{18}\text{Mg}_3\text{Ti}_2$ modulates the local solute availability and supersaturation to influence nucleation conditions, while Al_3Ti refines the microstructure to enhance diffusion kinetics and promote homogeneous nucleation. This combined effect creates a favorable environment for the abundant and dispersed precipitation of the β'' strengthening phase.

Fractographic observation also shows that the fracture of the 6061 joint exhibits relatively deep, small, and densely distributed dimples, which are characteristic of ductile fracture, and no pore-related defects are observed. By contrast, the fracture of the 4043 joint exhibits a large number of shallow dimples and pore-induced inclusions, indicating that defects dominate the fracture process. These defects act as stress concentrators, thus reducing the strength and ductility of the joint.

It is worth noting that the enhanced plasticity of the 6061 joints after PWHT also arises from the various strengthening mechanisms provided by TiC nanoparticles [89,90]. Specifically, TiC nanoparticles act as effective dislocation barriers, thereby increasing the material's resistance to deformation via pronounced dislocation pinning. Additionally, TiC at grain and phase boundaries strengthens interfacial bonding and promotes a more uniform distribution of stress. It also suppresses the initiation and early propagation of microcracks. During crack propagation, TiC and second-phase particles synergistically pin the grain boundaries. This impedes intergranular crack growth and redirects the crack path toward the HAZ. This process effectively reduces the crack-propagation rate and thereby improves the overall ductility of the material.

4.3. Assessment of strengthening mechanisms and synergy

To quantitatively elucidate the primary contributors to the yield strength evolution of the joints, the individual contributions from grain refinement strengthening and the strength recovery increment induced by PWHT were estimated separately. For the heat-treatable 6061 aluminum alloy, the formation of nanoscale β'' precipitates during PWHT is key to strength recovery and

enhancement. Other contributions (e.g., solid solution strengthening, dislocation strengthening) are not itemized but are collectively included in the experimentally measured yield strength.

(1) Contribution from Grain Refinement

The incremental yield strength contribution from grain refinement was calculated using the differential form of the Hall–Petch relationship:

$$\Delta\sigma_{HP} = k_y \left(\frac{1}{\sqrt{D_{WZ}^{6061}}} - \frac{1}{\sqrt{D_{WZ}^{4043}}} \right) \quad (2)$$

where $\Delta\sigma_{HP}$ is the yield strength increment due to grain refinement, k_y is the Hall–Petch slope (taken as $70\text{MPa}\mu\text{m}^{1/2}$ [91]), and D_{WZ} is the average grain size of the WZ obtained from EBSD results (Fig. 12, Fig. 13). The calculations indicate that the $\Delta\sigma_{HP}$ values contributed by grain refinement are 11.2MPa and 12.6MPa for the as-welded and PWHT conditions, respectively. This demonstrates that the direct contribution from grain refinement is on the order of $\approx 10\text{MPa}$.

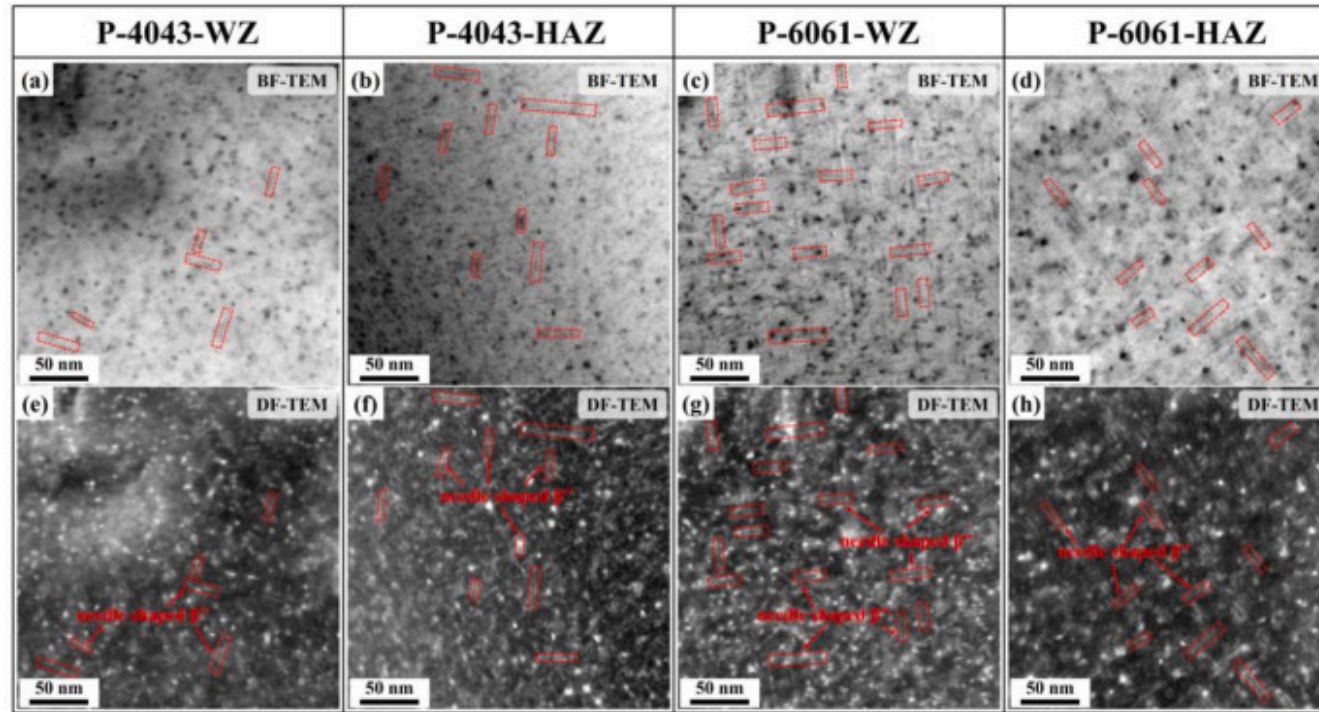
(2) Strength Recovery Increment Induced by PWHT

The difference in yield strength before and after PWHT characterizes the comprehensive strength recovery induced by the heat treatment:

$$\Delta\sigma_{PWHT} = \sigma_y^{PWHT} - \sigma_y \quad (3)$$

where σ_y^{PWHT} and σ_y are the yield strengths of the joint in the PWHT and as-welded conditions, respectively. For a given WZ, the change in grain size due to PWHT has a limited effect on strength. Therefore, $\Delta\sigma_{PWHT}$ can be considered to originate primarily from precipitation strengthening, while potentially including effects from dislocation recovery and residual stress relief. Based on the experimental measurements: for the 6061 joint, $\Delta\sigma_{PWHT} = 202 - 142.1 = 59.9\text{MPa}$; for the 4043 joint, $\Delta\sigma_{PWHT} = 204.6 - 137 = 67.6\text{MPa}$. Evidently, the PWHT-related increment is approximately 60–70MPa, constituting the dominant source of strength enhancement. TEM observations (Fig. 15) reveal a higher number density and more uniform distribution of nanoscale precipitates in the P-6061-WZ after PWHT, providing microstructural evidence supporting the “precipitation-dominant” conclusion.

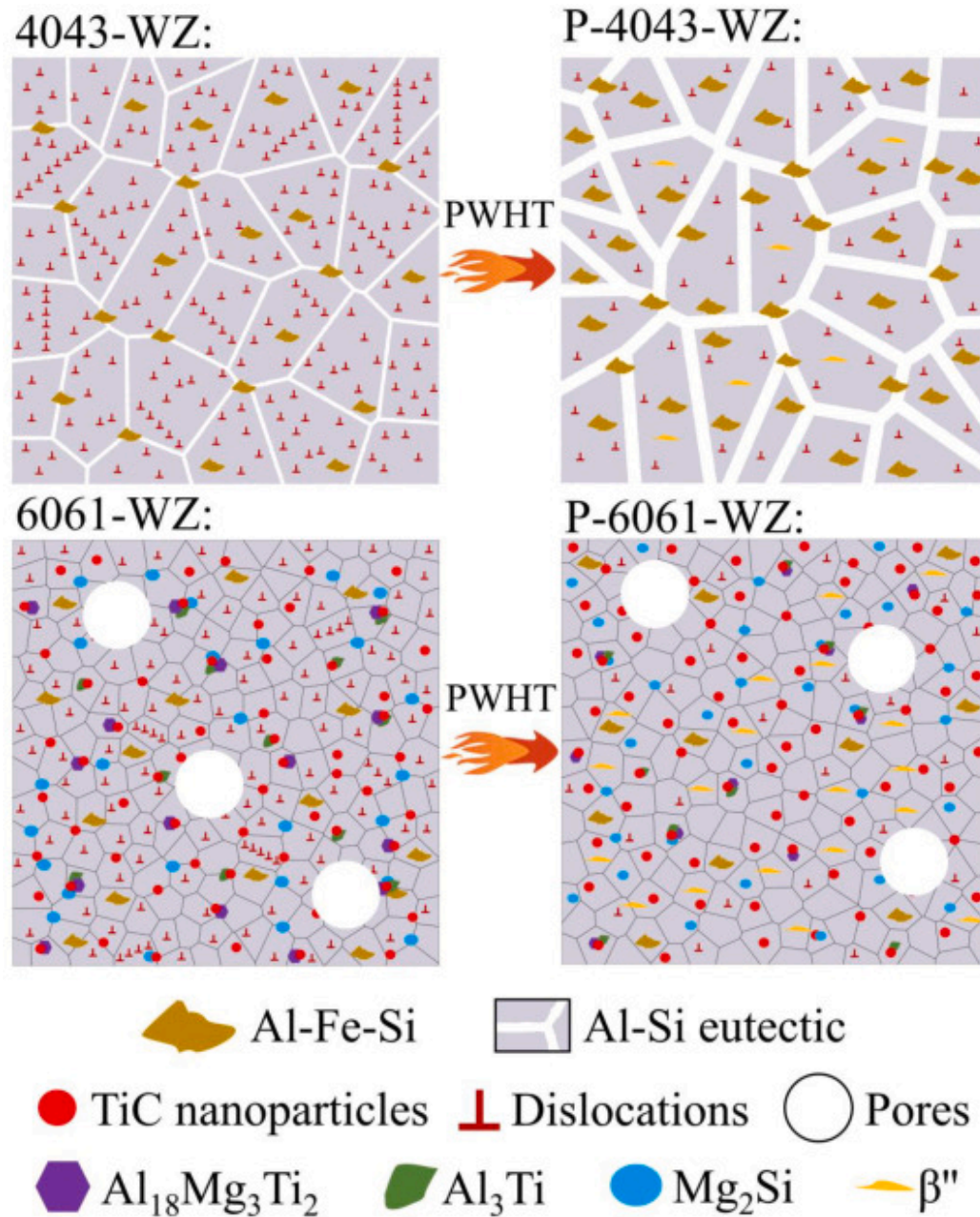
(3) Synergistic Effect and Porosity Influence



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Fig. 15. TEM analysis of fine precipitates in different regions under the PWHT condition: (a) P-4043-WZ; (b) P-4043-HAZ; (c) P-6061-WZ; (d) P-6061-HAZ.



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Fig. 16. Schematic of the microstructural evolution.

Synthesizing the above estimations, both grain refinement strengthening ($\approx 10\text{MPa}$) and age strengthening from PWHT ($\approx 60\text{--}70\text{MPa}$) collectively determine the yield strength level of the joint, with the latter being dominant. It is noteworthy that the role of TiC extends beyond direct grain refinement. The refined microstructure optimizes the spatial distribution of solute atoms, thereby promoting the formation of finer and more uniformly distributed nanoscale precipitates during subsequent PWHT (Fig. 15). This allows the precipitation strengthening contribution to be more fully realized within the refined structure, creating a positive synergistic gain between the two mechanisms.

On the other hand, welding porosity reduces the effective load-bearing area and compromises the macroscopic strength. Its influence can be described trend-wise using an empirical relationship [92]:

$$\sigma_{y,porous} = \sigma_{y,ideal}(1 - p)^2 \quad (4)$$

where $\sigma_{y,porous}$ is the yield strength of the joint with porosity, $\sigma_{y,ideal}$ is the yield strength of an ideal, defect-free joint, and p is the porosity fraction. Consequently, the final strength of the joint results from a balance between the enhancement from “grain refinement and precipitation strengthening” and the degradation caused by porosity defects. This also explains why, in some cases, a higher porosity fraction may constrain the macroscopic mechanical performance despite advantageous microstructural characteristics (finer grain size, higher precipitate density).

4.4. Porosity in 6061-WZ: Mechanisms and control

Under identical welding parameters, the porosity level in the 6061-WZ was significantly higher than that in the 4043-WZ. This discrepancy is primarily attributed to the influence of the introduced TiC nanoparticles on the flow and mass transfer behavior of the weld pool. The high specific surface area of the TiC nanoparticles is expected to increase the effective viscosity of the molten pool and suppress convective intensity, creating a “high-viscosity, weak-flow” environment [93]. Such an environment not only hinders the coalescence, growth, and upward escape of gas bubbles but may also increase the likelihood of bubbles being trapped by the advancing solidification front. Furthermore, TiC nanoparticles surfaces may adsorb hydrogen or carry oxide films, thereby providing potential heterogeneous nucleation sites for bubbles. The attachment of bubbles to these particles could further impede their mobility and escape capability. Consequently, the welding parameters, which were pre-set based on the fluid dynamic characteristics of a 4043 filler wire weld pool, were inadequate to ensure effective gas expulsion from the 6061–

TiC weld pool, leading to pore formation. This analysis indicates that the porosity issue in the 6061-WZ results from the combined effect of the intrinsic rheological changes in the weld pool induced by the TiC nanoparticles and the suboptimal welding parameters.

Despite non-optimized parameters, the 6061 joint still exhibited significant grain refinement and excellent resistance to hot cracking. The 6061 joints exhibit enhanced mechanical properties, particularly improved ductility after PWHT. Multiple strengthening mechanisms—grain refinement and precipitation strengthening—operate synergistically. These mechanisms provide a robust basis for achieving high strength and ductility in the similar-metal welding of 6000-series alloys. The mechanical properties obtained under the current conditions represent a combined outcome of the strengthening effects of TiC nanoparticles and the degradation caused by porosity defects. To quantitatively isolate the independent strengthening contribution of TiC, it is essential to fabricate a near-pore-free 6061-WZ for a direct comparison with a counterpart 4043-WZ of equivalent quality. Therefore, future process optimization for the 6061–TiC filler wire must be specifically designed to address the unique rheological behavior of this composite weld pool, aiming to enhance effective internal flow [[94], [95], [96]]. Strategies such as employing pulsed welding or torch weaving to periodically agitate the weld pool, or introducing electromagnetic stirring to impose forced convection, are promising active flow-field control techniques worthy of exploration. Concurrently, measures like rigorous cleaning of the filler wire and groove surfaces, along with appropriate preheating of the BM, can help reduce hydrogen sources and bubble nucleation probability, thereby facilitating the fabrication of high-quality WZs.

5. Conclusions

This work demonstrates a viable similar-metal welding strategy for 6061 aluminum alloy that overcomes the performance and recyclability limitations of mismatched filler wire approaches. This method integrates a TiC nanoparticle-reinforced 6061–TiC filler wire with a PWHT procedure. The results establish a promising pathway for sustainable and high-performance manufacturing of 6000-series alloys. The main findings are summarized as follows.

- (1) TiC nanoparticles within the WZ induced a microstructural transition in the as-welded state. They transformed coarse columnar dendrites ($83.4\mu\text{m}$) into refined equiaxed grains ($13.7\mu\text{m}$). This pronounced grain refinement simultaneously improved the solidified microstructure and enhanced resistance to solidification cracking.

- (2) The 6061 joints demonstrated markedly enhanced mechanical performance in both the as-welded and PWHT states. After PWHT, they achieved a tensile strength of 281.7 MPa and an elongation of 17.0%, corresponding to a 59% improvement in ductility over the P-4043 joints. The fracture surface exhibited a typical ductile morphology with fine, uniformly distributed dimples, consistent with the material's high ductility.
- (3) The P-6061 joint exhibits excellent overall mechanical properties. The improvement in performance results from the use of 6061–TiC filler wire, and a synergistic strengthening mechanism, involving nanoparticle-induced solidification refinement and PWHT-controlled precipitation phase coupling, is established. TiC nanoparticles first refine the grains and inhibit crack initiation and propagation, thereby improving plasticity. Subsequent PWHT promotes the precipitation of a nano-scale β'' strengthening phase, significantly enhances the strength, and achieves an excellent strength–plasticity balance.
- (4) PWHT enhances the hardness and strength of both joints. However, owing to the Al–Mg–Si composition of the 6061–WZ, which can undergo precipitation strengthening, together with TiC-induced grain refinement and a nearly random weak texture, the 6061 joint exhibits both high strength and uniform plastic deformation. These microstructural features thereby enhance the joint's ductility. By contrast, the compositional mismatch in 4043–WZ leads to the formation of brittle Al–Si eutectic, coarse Si-rich particles and a strong texture, as well as an insufficient amount of the β'' strengthening phase. This promotes strain localization and crack initiation, thereby limiting the plasticity of the 4043 joint.
- (5) The presence of porosity, attributable to non-optimized welding parameters, did not compromise the pronounced grain refinement and robust hot-cracking resistance achieved in the 6061 joints. This result highlights a clear pathway for future enhancement: optimizing welding parameters to minimize porosity is expected to further improve mechanical performance.

CRedit authorship contribution statement

Wenbin Jiang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation.

Chenxu Zhao: Supervision, Methodology, Investigation. **Yao Huang:** Supervision, Methodology, Investigation. **Yanjian Ding:**

Visualization. **Liang Qi:** Writing – review & editing, Supervision, Formal analysis. **Jiqiang Chen:** Writing – review & editing, Supervision, Formal analysis. **Zhou Li:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors are grateful for The Doctoral Scientific Research Foundation of Jiangxi University of Science and Technology (Grant no. [205200100693](#)), China Postdoctoral Science Foundation (Grant no. [2023M740350](#)) and Program of Qingjiang Excellent Young Talents of Jiangxi University of Science and Technology (Grant no. [JXUSTQJYX2020022](#)).

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Data availability

Data will be made available on request.

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